Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 97
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A developer has designed an application to store incoming data as JSON files in Amazon S3 objects. Custom business logic in an AWS Lambda function then transforms the objects, and the Lambda function loads the data into an Amazon DynamoDB table. Recently, the workload has experienced sudden and significant changes in traffic. The flow of data to the DynamoDB table is becoming throttled.

The developer needs to implement a solution to eliminate the throttling and load the data into the DynamoDB table more consistently.

Which solution will meet these requirements?

A. Refactor the Lambda function into two functions. Configure one function to transform the data and one function to load the data into the DynamoDB table. Create an Amazon Simple Queue Service (Amazon SQS) queue in between the functions to hold the items as messages and to invoke the second function.

B. Turn on auto scaling for the DynamoDB table. Use Amazon CloudWatch to monitor the table's read and write capacity metrics and to track consumed capacity.

C. Create an alias for the Lambda function. Configure provisioned concurrency for the application to use.

D. Refactor the Lambda function into two functions. Configure one function to store the data in the DynamoDB table. Configure the second function to process the data and update the items after the data is stored in DynamoDB. Create a DynamoDB stream to invoke the second function after the data is stored.

Show Suggested Answer

by MrTee at *April 21, 2023, 9:02 p.m.*

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Comments

ihebchorfi Highly Voted 2 years, 11 months ago

Selected Answer: A

A. Refactor the Lambda function into two functions. Configure one function to transform the data and one function to load the data into the DynamoDB table. Create an Amazon Simple Queue Service (Amazon SQS) queue in between the functions to hold the items as messages and to invoke the second function.

By breaking the Lambda function into two separate functions and using an SQS queue to hold the transformed data as messages, you can decouple the data transformation and loading processes. This allows for more controlled loading of data into the DynamoDB table and helps eliminate throttling issues.

upvoted 26 times

MrTee Highly Voted 2 years, 11 months ago

Selected Answer: D

This solution will allow the developer to store the incoming data into the DynamoDB table more consistently without being throttled. By splitting the Lambda function into two functions, the first function can store the data into the DynamoDB table and exit quickly, avoiding any throttling issues. The second function can then process the data and update the items after the data is stored in DynamoDB using a DynamoDB stream to invoke the second function.
Option A is also a good option but not the best solution because it introduces additional complexity and cost by using an Amazon SQS queue.

upvoted 9 times

Ashwinvdm22 2 years, 2 months ago

The problem I have with option D is that it is adding more lad on the DynamoDB table. What is the need to insert the item and then update the item later. This is performing two operation on every item just to get it into the correct state. I would go with option A since it is not performing two operations on the DB and hence reducing the load which will help with throttling.

upvoted 1 times

robotgeek 2 years, 10 months ago

Sorry but when you say "the first function can store the data into the DynamoDB table and exit quickly, avoiding any throttling issues" I dont understand your point

upvoted 5 times

[Removed] 2 years, 3 months ago

The issue is between S3 to DynamoDB this is where we need to fix the bottleneck. So configuring two functions to work on the data after it has been uploaded to DynamoDB makes no sense.

upvoted 1 times

nbxyzd 1 year, 5 months ago

Read carefully: The flow of data to the DynamoDB table is becoming throttled.
So the bottleneck is the DynamoDB, not the lambda function transforming the data. Option D doesn't help because the first function storing data into the DynamoDB will still hit the throttling issue.

upvoted 2 times

mooncake1 Most Recent 1 year, 2 months ago

Selected Answer: B

Adjusting the WCU is enough. no need to implement such complex solutions

upvoted 1 times

sumanshu 1 year, 3 months ago

Selected Answer: A

The SQS queue acts as a buffer, which smooths out sudden traffic spikes by queuing up data. The second Lambda function processes data from the queue at a steady rate, reducing the likelihood of throttling in DynamoDB.

B) Eliminated - Auto scaling takes time to adjust, during which throttling may still occur.

C) Eliminated - Provisioned concurrency ensures that the Lambda function can handle a predictable number of concurrent requests, but it does not solve the throttling issue at the DynamoDB layer

upvoted 2 times

nbxyzd 1 year, 5 months ago

Selected Answer: A

It's hands-down A.

upvoted 1 times

Tluszczyk 1 year, 8 months ago

Selected Answer: B

"A" would be optimal, but without backoff algorithm the lambda division and SQS won't affect the throttling. However Dynamo can autoscale https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/AutoScaling.html

upvoted 1 times

queekao 1 year, 8 months ago

Yes, but it asks to "load the data into the DynamoDB table more consistently." Therefore, option A will prevent unintentional data load into DynamoDB, it's the best option.

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: A

A is the correct answer.

upvoted 1 times

nder 2 years ago

Selected Answer: B

we are trying to stop throttling...

upvoted 1 times

SerialiDr 2 years, 1 month ago

Selected Answer: A

This solution addresses the need to eliminate throttling and ensure consistent data loading into the Amazon DynamoDB table by separating the transformation and loading processes into two different functions. Using an Amazon SQS queue to hold items as messages between the two functions helps manage the flow of data and prevents overloading the DynamoDB table, thereby eliminating throttling issues.

upvoted 2 times

Brisun 2 years, 2 months ago

Selected Answer: A

A is correct as it requires to write to DynamoDB "more consistently". Option B can solve the problem too but the writing won't be consistent as the traffic will go up and down instantly.

In reality, I will probably do Option B only.

upvoted 3 times

SD_CS 2 years, 2 months ago

Selected Answer: B

I do not feel refactoring the data transformation and loading would help here as I do not think the number of concurrent calls to the DB would decrease because of this. Autoscaling DynamoDB would seem a more potent option to me.

upvoted 4 times

peekingpicker 2 years, 2 months ago

Selected Answer: B

Why not B ?
DynamoDB can autoscale the RCU and WCU

upvoted 4 times

SerialiDr 2 years, 2 months ago

Selected Answer: A

A. Refactor the Lambda function into two functions, using an Amazon SQS queue to manage the data flow, and/or

B. Turn on auto scaling for the DynamoDB table to automatically adjust its write capacity based on traffic patterns.

Both A and B address the core issue of managing write throughput to the DynamoDB table to prevent throttling. Option A provides a way to smooth out data flow and manage write requests more effectively, while option B allows the table to scale its capacity automatically in response to changing traffic, although with potential limitations in response speed to sudden traffic spikes. Combining these approaches could provide an even more robust solution.

upvoted 3 times

KarBiswa 2 years, 3 months ago

Selected Answer: A

Off course A & D are options but here after inserting the data further we cannot modify because one extra writing cost will incur rather using queue lambda can poll the transformed data

upvoted 3 times

Nagasoracle 2 years, 5 months ago

Selected Answer: A

Answer : A
SQS can be configured to invoke Lambda.
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-configure-lambda-function-trigger.html

upvoted 4 times

dexdinh91 2 years, 5 months ago

Selected Answer: B

I think B

upvoted 1 times

jingle4944 2 years, 5 months ago

Lambda functions can be triggered by SQS:
https://docs.aws.amazon.com/AWSSimpleQueueService/latest/SQSDeveloperGuide/sqs-configure-lambda-function-trigger.html

upvoted 2 times

nbxyzd 1 year, 5 months ago

Correct, and most importantly, it's triggered *synchronously* so that there won't be throttling issue. Quote:
You can use an AWS Lambda function to process messages in an Amazon SQS queue. Lambda polls the queue and invokes your Lambda function synchronously with an event that contains queue messages.

upvoted 1 times